IGCSE 0607 Practice Questions — E4.5 — Combined Questions & Markscheme

Parallel lines

Total Questions: 9 — **Total Marks:** 44 Marks

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- **E4.5** Parallel lines
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

1 Find the equation of the line that is parallel to $y = -2x + 7$ and passes through the point $(3, 4)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

2 Line L has the equation $4x + 2y = 9$.
Find the equation of the line parallel to L that passes through the point $(3, -1)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

3 Line L has the equation $y = 6 - 3x$.

Find the equation of the line parallel to L that passes through the point $(-2, 5)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

4 The points A and B have coordinates $(-3, 5)$ and $(5, -1)$ respectively.
Line L is parallel to the line segment AB and passes through the point $(4, -2)$.

(a) Find the gradient of the line segment AB.

..... **[2]**

(b) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

5 Line M has the equation $5x - 2y = 15$.
Point P has coordinates $(-4, 1)$.
Find the equation of the line parallel to M that passes through P.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

6 A is the point $(-3, 2)$ and B is the point $(5, 6)$.

(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Line L is parallel to AB and passes through the point $(4, -1)$.
Find the equation of line L, giving your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

7 Points $A(-1, 5)$ and $B(3, -3)$ lie on line L_1.
Line L_2 is parallel to L_1 and passes through the point $C(2, 7)$.
Line L_2 crosses the x-axis at D and the y-axis at E.

(a) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(b) Calculate the exact area of triangle ODE, where O is the origin $(0, 0)$.

..... $units^2$ **[3]**

8 Line L_1 has the equation $3x - 4y = 12$.
Line L_2 is parallel to L_1.

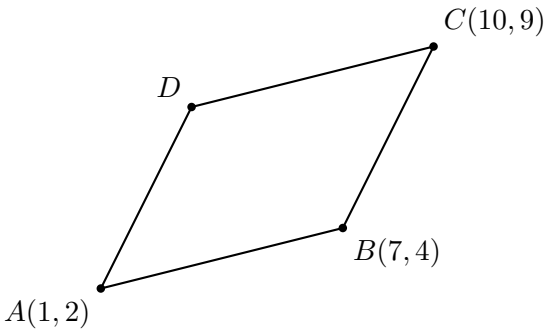
Line L_2 crosses the x-axis at point A and the y-axis at point B.
The length of the line segment AB is 15.

Find the two possible equations of L_2.
Give your answers in the form $y = mx + c$.

.....

..... **[7]**

9 The points $A(1,2)$, $B(7,4)$, and $C(10,9)$ are three vertices of a parallelogram $ABCD$.
Line CD is parallel to line AB and line AD is parallel to line BC.



Not to scale

(a) Find the equation of the line CD.

..... **[3]**

(b) Find the equation of the line AD.

..... **[3]**

(c) Work out the coordinates of the vertex D.

(..... , ...) **[3]**

1 Identifies the gradient $m = -2$ and sets up $4 = -2(3) + c$ or $y - 4 = -2(x - 3)$ **[M1]**

$-2x + 10$ **[A1]**

[Back to Question] *[Total Marks: 2]*

2 Rearranges equation to find gradient, e.g., $2y = -4x + 9$ implies $m = -2$ **[M1]**

Substitutes $(3, -1)$ into $y = -2x + c$ or writes $y - (-1) = -2(x - 3)$ **[M1]**

$-2x + 5$ **[A1]**

[Back to Question] *[Total Marks: 3]*

3 Gradient $m = -3$ **[B1]**

$5 = -3(-2) + c$ or $y - 5 = -3(x - (-2))$ **[M1]**

$y = -3x - 1$ **[A1]**

[Back to Question] *[Total Marks: 3]*

4 (a) $\dfrac{-1-5}{5-(-3)}$ or $\dfrac{5-(-1)}{-3-5}$ **[M1]**

$-\dfrac{3}{4}$ **[A1]**

(b) $m = -\dfrac{3}{4}$ **[M1]**

$-2 = -\dfrac{3}{4}(4) + c$ or $y - (-2) = -\dfrac{3}{4}(x - 4)$ **[M1]**

$y = -\dfrac{3}{4}x + 1$ **[A1]**

[Back to Question] *[Total Marks: 5]*

5 Rearranges $5x - 2y = 15$ to find the gradient, e.g. $2y = 5x - 15$

y $= \dfrac{5}{2}x - 7.5$, so $m = \dfrac{5}{2}$ [M1]

Substitutes $(-4, 1)$ and $m = \dfrac{5}{2}$ into $y = mx + c$: $1 = \dfrac{5}{2}(-4) + c$ **[M1]**

$\dfrac{5}{2}x + 11$ or $2.5x + 11$ **[A1]**

[Back to Question] *[Total Marks: 3]*

6 (a) $\dfrac{6-2}{5-(-3)}$ oe **[M1]**

$\dfrac{1}{2}$ or 0.5 **[A1]**

(b) $y-(-1)=\frac{1}{2}(x-4)$ oe **[M1]**

$2(y+1)=x-4$ or $2y+2=x-4$ **[M1]**

$x-2y=6$ or $-x+2y=-6$ **[A1]**

[Back to Question] *[Total Marks: 5]*

7 (a) Calculates the gradient of L_1: $m=\frac{-3-5}{3-(-1)}$ **[M1]**

$m=\frac{-8}{4}=-2$ **[A1]**

Uses parallel gradient -2 and point $C(2,7)$ to find c: $7=-2(2)+c$ **[M1]**

$-2x+11$ **[A1]**

(b) Finds y-intercept $E(0,11)$ and sets $y=0$ to find x-intercept D: $0=-2x+11$

x = 5.5[$M1$]

Sets up area calculation for right-angled triangle: $\frac{1}{2}\times 5.5\times 11$ **[M1]**

30.25 or $\frac{121}{4}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

8 Gradient of $L_1=\frac{3}{4}$ oe **[B1]**

Equation of L_2 is $y=\frac{3}{4}x+c$ or $3x-4y=k$ **[M1]**

x-intercept of L_2 is $-\frac{4c}{3}$ (or $\frac{k}{3}$) AND y-intercept is c (or $-\frac{k}{4}$) oe **[M1]**

$(-\frac{4c}{3})^2+c^2=15^2$ or $(\frac{k}{3})^2+(-\frac{k}{4})^2=15^2$ oe **[M1]**

$\frac{25c^2}{9}=225$ or $\frac{25k^2}{144}=225$ **[M1]**

$c=\pm 9$ or $k=\pm 36$ **[A1]**

$y=\frac{3}{4}x+9$ and $y=\frac{3}{4}x-9$ **[A1]**

[Back to Question] *[Total Marks: 7]*

9 (a) Gradient of $AB=\frac{4-2}{7-1}=\frac{1}{3}$ **[M1]**

Substitute $(10,9)$ into $y-y_1=m(x-x_1)$ with their gradient: $y-9=\frac{1}{3}(x-10)$ **[M1]**

$y=\frac{1}{3}x+\frac{17}{3}$ or $3y=x+17$ **[A1]**

(b) Gradient of $BC=\frac{9-4}{10-7}=\frac{5}{3}$ **[M1]**

Substitute $(1, 2)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 2 = \frac{5}{3}(x - 1)$ **[M1]**

$y = \frac{5}{3}x + \frac{1}{3}$ or $3y = 5x + 1$ **[A1]**

(c) Equate equations from (a) and (b): $\frac{1}{3}x + \frac{17}{3} = \frac{5}{3}x + \frac{1}{3}$ **[M1]**

$4x = 16 \implies x = 4$ **[A1]**

$y = 7$ **[A1]**

[Back to Question] *[Total Marks: 9]*